

Overview of the Bioinformatics Workflow

PHASE 1 – DATA ACQUISITION



Raw FASTQ Files
from ENA / NCBI Database

PHASE 2 – DATA PREPROCESSING



Quality Control
(FastQC)



Quality Summary
(MultiQC)

*If reads are
low quality*



Read Trimming
(Trimmomatic)

*If reads are
of acceptable
quality*



Read Mapping
(BWA-MEM)



BAM Processing
(SAMtools)



Variant Calling
(BCFtools + iVar)



Variant Filtering

PHASE 3 – DOWNSTREAM ANALYSIS



Functional Annotation
(SnpEff)



Gene Ontology (GO)
Enrichment Analysis



KEGG Pathway
Enrichment Analysis



Clustering Analysis
(R and Python)



Biological Interpretation